Luke Auderer

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Education

Iowa State University
Ames, IA

Master of Science in Artificial Intelligence — *Aug. 2025 - May 2027 (expected)*

- GPA: 4.0/4.0

Bachelor of Science in Computer Engineering — *Aug. 2023 - May 2026*

- GPA: 4.0/4.0, President's List, Engineering Dean's List

Des Moines Area Community College
Ankeny, IA

Associate of Science and Associate of Arts (earned in high school) — *Aug. 2019 - May 2023*

- GPA: 4.0/4.0, ACT Score: 35/36, Iowa Governor's Scholar Award, Iowa Seal of Biliteracy (Spanish proficiency)

Experience

Google
Seattle, WA

Machine Learning Intern — *May 2026 - Aug. 2026 (expected)*

- Optimizing LLM inference on Google's latest-generation TPUs through system and model-level performance improvements
- Developing an agentic workflow that automatically ports newly supported vLLM models and CUDA kernels to JAX/Pallas for high-performance TPU execution, reducing model bring-up time from weeks to days

SpaceX
Austin, TX

Software Engineering Intern — *June 2025 - Aug. 2025*

- Developed firmware for Starlink dishes and power supply units to collect and aggregate power telemetry across the global Starlink fleet, visualizing the results in a Grafana dashboard for real-time monitoring
- Integrated the telemetry data into Starlink's test infrastructure, enabling engineers to track performance trends and perform root-cause analysis of connectivity issues during testing
- Identified and resolved a major recurring failure source through the collected telemetry insights, saving an estimated $35,000 per week

John Deere
Urbandale and Ankeny, IA

CVML Robotics Engineer (Full-Year Intern) — *July 2024 - May 2026*

- Trained an image classification model to characterize images from various environmental conditions on a John Deere planter in real time to provide signals for automation
- Supported a large-scale data aggregation and labeling effort to build a high-quality training dataset for the model
- Developed software for an embedded camera system and machine vision controller, helping transition it to a customer-ready product
- Improved the functionality and UX design of the display application for the embedded camera system
- Created and updated a suite of hardware integration tests in Python to verify hardware communication and functionality

Embedded Software Engineer (Full-Year Intern) — *May 2023 - July 2024*

- Developed embedded software for John Deere sprayers to enable real-time communication between vehicle subsystems over the CAN bus
- Diagnosed and corrected defects in the nozzle control algorithm for John Deere sprayers using MATLAB/Simulink
- Updated the Qt-based display application used in John Deere sprayers with new UI components and pages to support additional features

Iowa State University Advanced Manufacturing Lab
Ames, IA

Research Assistant - Project Lead — *Sep. 2023 - Present*

- Leading a project under Dr. Frank Peters to develop a structured-light 3D scanner as a low-cost alternative to high-end commercial products
- Delivered a prototype scanner under a $2,000 budget that achieves a scan accuracy within 4% of high-end models costing over $20,000
- Designed and built the prototype integrating electronics, 3D-printed parts, machine vision cameras, and a digital light projector to achieve synchronized image capture and high-quality 3D scans
- Optimized image processing workflows to enhance the speed/precision of 3D scans using OpenCV and open-source scanning software

Activities

FIRST Robotics
Polk City, IA

Programming Team, Current Volunteer — *June 2021 - Present*

- Built a swerve drivetrain for the team robot to improve handling, then integrated an onboard camera system for autonomous navigation
- Returned to my high school team as a volunteer to help build the robot practice field each year and assist with other team projects

Technical Skills

Languages: Python, Java, C++, C, JavaScript (React.js, Node.js), HTML/CSS, MATLAB, SQL
Tools: Git, Linux, Docker, OpenCV, PyTorch, CUDA, JAX/Pallas, vLLM, MLflow, Databricks, AWS, Voxel51, Ansible, Grafana, Qt, CMake, ROS